Ryan Zheng

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Education

University of California, Los Angeles — Los Angeles, CA
B.S. in Physics, Data Science Engineering Minor; ***GPA: 3.875*** — *Sep. 2024 – June 2028*
Relevant Coursework: Probability & Stats, Linear Algebra, Differential Equations, Discrete Math, Machine Learning

Awards & Publications

USA Physics Olympiad (USAPhO) Semifinalist (2023) — **top 8%** of 5,173 competitors

IMC Prosperity 4 (2026) — **top 8%** of 18,803 teams

ACM ICPC Break the Binary Medalist (2025) — **3rd place** of 25 teams

Sigma Pi Sigma Physics Honor Society · ACM AI Projects Director · Jane Street Puzzle Solver

Chen, Y., Jiao, J., & Zheng, R. (2024). Trip generation and built environment impacts: regular vs. essential trips in the U.S. *Proceedings of CICTP 2024*, 3317–3326.

Experience

Incoming Quant Trading Intern — June 2027
IMC Trading — *Chicago, IL*

Software Engineer Intern — June 2026 – September 2026
Amazon — *Seattle, WA*

- Architect a distributed ML experiment-orchestration platform (AWS Step Functions) that compiles declarative configs into fault-tolerant DAGs of multi-node GPU workloads, running unattended multi-day pipelines.
- Design order-independent checkpoint-selection algorithm and parallelize evaluation fan-out to overlap with in-flight training; cut end-to-end runtime by **>50%** and inter-stage handoff latency from **hours to seconds**.
- Build config compiler that deterministically deduplicates state machines, cut experiment setup from **∼30 min to <1 min**, streamline research iteration with agentic failure recovery.

Machine Learning Researcher — January 2026 – present
UCLA Scalable Analytics Institute — *Los Angeles, CA*

- Restructure **3.9 TB** PET/CT archive into a longitudinal panel of **80 subjects** × **4 timepoints**, and evaluate 4 pretrained encoders. Forecast subject-level embeddings one timestep ahead under leave-one-subject-out CV, beating a random-walk benchmark on 66% of held-out transitions while avoiding look-ahead bias.
- Build vision-language model, injecting scan and forecast embeddings as image tokens into a LoRA-finetuned 8B language model with joint regression and classification heads; predict disease outcome **8 weeks ahead** at **0.76 AUC / 72%** vs. **0.58 / 50%** scan-only under matched-budget ablation (10 LOSO runs). Submitting to ICLR.

Machine Learning Intern — January 2025 – January 2026
Scale AI — *San Francisco, CA (Remote)*

- Built statistical evaluation rubrics and Dockerized agentic RL evaluation environments, supporting industry-standard AI benchmarks (e.g., Aider LLM Leaderboards) across **200+ datapoints** and **6 clients**.

Quantitative Projects

Perpetual Futures Funding Carry Strategy | *Personal Project* — May 2026 – Present

- Built a C++ market-data simulation pipeline processing **400M+ order-book events** across 12 months of BTC/SOL data, reconstructing historical books and modeling execution costs and slippage at varying order sizes.
- Designed a decoupled ingestion and simulation architecture using a **lock-free queue**, enabling high-throughput evaluation of a funding-rate carry strategy; measured **0.8 Sharpe** net of costs and **$210K** executable capacity.

S&P500 Return Forecasting | *ACM AI, Kaggle Competition* — October 2025 – December 2025

- Ensembled daily-horizon forecasting models for S&P500 forward returns using gradient-boosted trees, feed-forward networks, and LSTMs. Achieved **1.48 Sharpe** over 6-month scoring period; **top 22%** of submissions.

Skills & Interests

Technical: C++, Java, Python, NumPy, statsmodels, pandas, SciPy, PyTorch, R, SQL, Linux
Languages: English (native), Mandarin (fluent)
Interests: Skiing, tennis, jazz piano, competitive archery, game speedrunning (record holder), poker